```
import pandas as pd
data={
    'Name':['A','B','C','D','E'],
    'Math':[47,89,56,90,76,],
    'Science':[90,78,67,89,56],
    'English':[66,89,67,45,90]
     }
df=pd.DataFrame(data)
print(df)
```

```
  Name  Math  Science  English
0    A    47       90       66
1    B    89       78       89
2    C    56       67       67
3    D    90       89       45
4    E    76       56       90
```

```
import numpy as np
marks=df[["Math","Science","English"]].values
average_marks=np.mean(marks,axis=1)
df["Average"]=average_marks
print(df)
```

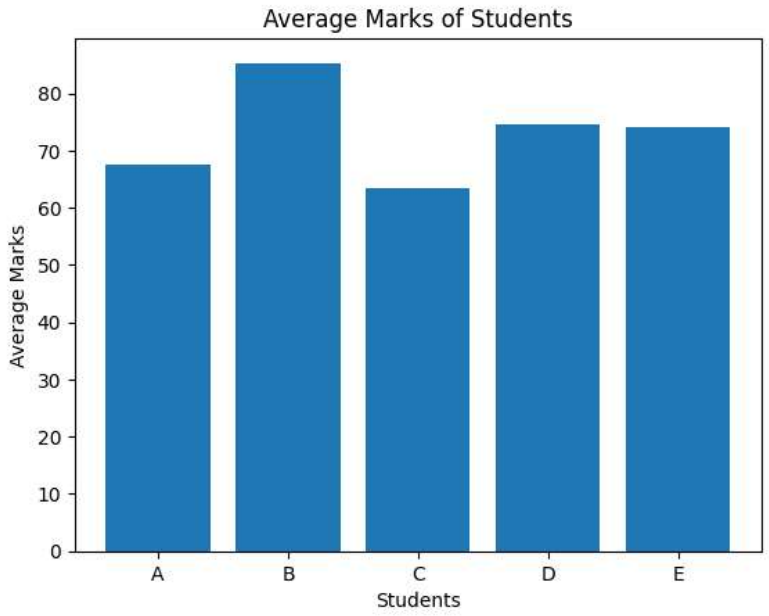
```
  Name  Math  Science  English    Average
0    A    47       90       66  67.666667
1    B    89       78       89  85.333333
2    C    56       67       67  63.333333
3    D    90       89       45  74.666667
4    E    76       56       90  74.000000
```

```
def grade(avg):
  if avg>=90:
    return 'A'
  elif avg>=75:
    return 'B'
  elif avg>=60:
    return 'C'
  else:
    return 'D'
df['Grade']=df["Average"].apply(grade)
df
```

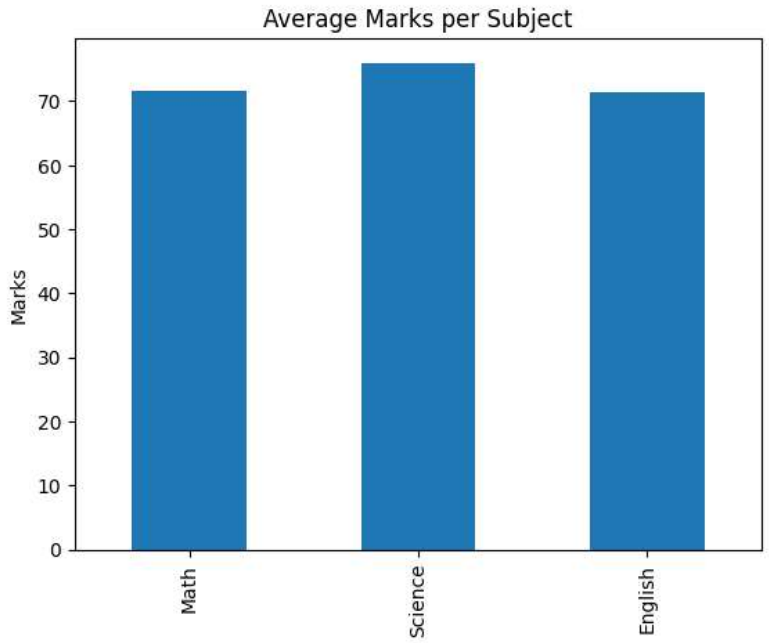
	Name	Math	Science	English	Average	Grade
0	A	47	90	66	67.666667	C
1	B	89	78	89	85.333333	B
2	C	56	67	67	63.333333	C
3	D	90	89	45	74.666667	C
4	E	76	56	90	74.000000	C

```
import matplotlib.pyplot as plt

plt.figure()
plt.bar(df["Name"], df["Average"])
plt.xlabel('Students')
plt.ylabel('Average Marks')
plt.title('Average Marks of Students')
plt.show()
```




```
import matplotlib.pyplot as plt
df[["Math","Science","English"]].mean().plot(kind="bar")
plt.title("Average Marks per Subject")
plt.ylabel("Marks")
plt.show()
```


Average Marks per Subject
Marks
0
10
20
30
40
50
60
70
Math
Science
English